

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-004 (W004) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W004
Well Name	SYN-Assam-Duliajan-004
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.412500 N / 95.349472 E
Rig	SYN-RIG-E1400-04
Spud Date	28-Oct-2008
Final Report Date	08-Nov-2008
Daily Reports in Set	12 (Day 01 to Day 12)
Total Depth Reached	3,590 m
Primary Formation	Sandstone
Events Reported in Set	2

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	28-Oct-2008	0	500	500	Claystone	—
02	29-Oct-2008	500	940	440	Claystone / Shale	—
03	30-Oct-2008	940	1,330	390	Shale	—
04	31-Oct-2008	1,330	1,685	355	Shale	—
05	01-Nov-2008	1,685	2,005	320	Shale	—
06	02-Nov-2008	2,005	2,300	295	Shale / Sandstone	—
07	03-Nov-2008	2,300	2,570	270	Sandstone	—
08	04-Nov-2008	2,570	2,825	255	Sandstone	—
09	05-Nov-2008	2,825	3,010	185	Sandstone	Mud Loss
10	06-Nov-2008	3,010	3,230	220	Sandstone / Limestone	—
11	07-Nov-2008	3,230	3,390	160	Limestone	Mud Loss
12	08-Nov-2008	3,390	3,590	200	Limestone	—

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). Depths, dates, parameters and events are generated from the prototype datasets and are not operational records of any company.

DOC No. DDR-W004-SET-R0 · PROTOTYPE TRAINING RECORD

DAILY DRILLING REPORT — DAY 01

W004 · 28-Oct-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-10-28
Day Number: 1
Start Depth: 0 m
End Depth: 500 m
Drilled Interval: 500 m
Formation: Claystone
Mud Weight: 9.37 ppg
ROP: 32.2 m/hr
RPM: 142
WOB: 7.1 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Rig up completed and well spudded at 28-Oct-2008. Drilled 17-1/2" hole from surface to 500 m.
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are claystone, consistent with the 0–822 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 9.37 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 500 m in Claystone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 02

W004 · 29-Oct-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-10-29
Day Number: 2
Start Depth: 500 m
End Depth: 940 m
Drilled Interval: 440 m
Formation: Claystone / Shale
Mud Weight: 10.05 ppg
ROP: 27.3 m/hr
RPM: 135
WOB: 11.2 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 440 m of section from 500 m to 940 m (16.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.
- Pulled out of hole, ran 13-3/8" casing to 820 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.

Geological observations.

Formation change crossed during the tour: Claystone to 822 m, Shale to 2,009 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 10.05 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 940 m in Claystone. 13-3/8" casing cemented at 820 m; waiting on cement.

DAILY DRILLING REPORT — DAY 03

W004 · 30-Oct-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-10-30
Day Number: 3
Start Depth: 940 m
End Depth: 1330 m
Drilled Interval: 390 m
Formation: Shale
Mud Weight: 10.73 ppg
ROP: 16.8 m/hr
RPM: 127
WOB: 15.9 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 390 m of Shale from 940 m to 1,330 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 822–2,009 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 10.73 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,330 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 04

W004 · 31-Oct-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-10-31
Day Number: 4
Start Depth: 1330 m
End Depth: 1685 m
Drilled Interval: 355 m
Formation: Shale
Mud Weight: 11.25 ppg
ROP: 16.3 m/hr
RPM: 123
WOB: 18.5 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 355 m of Shale from 1,330 m to 1,685 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 822–2,009 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.25 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,685 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 05

W004 · 01-Nov-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-11-01
Day Number: 5
Start Depth: 1685 m
End Depth: 2005 m
Drilled Interval: 320 m
Formation: Shale
Mud Weight: 11.73 ppg
ROP: 15.2 m/hr
RPM: 120
WOB: 21.2 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 320 m of Shale from 1,685 m to 2,005 m (22.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 822–2,009 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.73 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,005 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 06

W004 · 02-Nov-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-11-02
Day Number: 6
Start Depth: 2005 m
End Depth: 2300 m
Drilled Interval: 295 m
Formation: Shale / Sandstone
Mud Weight: 11.97 ppg
ROP: 17.6 m/hr
RPM: 117
WOB: 22.5 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 295 m of section from 2,005 m to 2,300 m (16.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.
- Pulled out of hole, ran 9-5/8" casing to 2,010 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 700 m). Waited on cement and pressure tested.

Geological observations.

Formation change crossed during the tour: Shale to 2,009 m, Sandstone to 3,180 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.97 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,300 m in Sandstone. 9-5/8" casing cemented at 2,010 m; waiting on cement.

DAILY DRILLING REPORT — DAY 07

W004 · 03-Nov-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-11-03
Day Number: 7
Start Depth: 2300 m
End Depth: 2570 m
Drilled Interval: 270 m
Formation: Sandstone
Mud Weight: 12.36 ppg
ROP: 15.8 m/hr
RPM: 113
WOB: 24.4 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 270 m of Sandstone from 2,300 m to 2,570 m (22.5 hr on bottom).
- Deviation survey taken; hole angle within tolerance.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,009–3,180 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.36 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,570 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 08

W004 · 04-Nov-2008

Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Report Date: 2008-11-04
Day Number: 8
Start Depth: 2570 m
End Depth: 2825 m
Drilled Interval: 255 m
Formation: Sandstone
Mud Weight: 12.74 ppg
ROP: 15.6 m/hr
RPM: 110
WOB: 26.3 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 255 m of Sandstone from 2,570 m to 2,825 m (22.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,009–3,180 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.74 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,825 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 09						W004 · 05-Nov-2008														
Well ID: W004 Well Name: SYN-Assam-Duliajan-004 Report Date: 2008-11-05 Day Number: 9 Start Depth: 2825 m End Depth: 3010 m Drilled Interval: 185 m Formation: Sandstone Mud Weight: 13.02 ppg ROP: 14.5 m/hr RPM: 110 WOB: 27.9 klbf Events: Mud Loss at 2865 m (Medium) NPT: 6.8 hours																				
Operations performed. - Drilled 185 m of Sandstone from 2,825 m to 3,010 m (15.7 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Mud Loss handled at 2,865 m — 6.8 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.																				
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,009–3,180 m unit logged in the offset wells of this cluster.																				
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 13.02 ppg after treatment.																				
Equipment / operational observations. Mud pump No. 2 packing changed during routine circulation; no lost time.																				
Events reported this day. <table><tr><th>Event ID</th><th>Depth (m)</th><th>Formation</th><th>Type</th><th>Severity</th><th>Hr</th><th>Operational Impact</th></tr><tr><td>E0004</td><td>2,865</td><td>Sandstone</td><td>Mud Loss</td><td>Medium</td><td>6.8</td><td>Partial non-productive time; drilling suspended while the well was treated and conditioned.</td></tr></table>							Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact	E0004	2,865	Sandstone	Mud Loss	Medium	6.8	Partial non-productive time; drilling suspended while the well was treated and conditioned.
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact														
E0004	2,865	Sandstone	Mud Loss	Medium	6.8	Partial non-productive time; drilling suspended while the well was treated and conditioned.														
End-of-day status. Depth 3,010 m in Sandstone. Drilling ahead. Non-productive time this tour 6.8 hr.																				

DAILY DRILLING REPORT — DAY 10						W004 · 06-Nov-2008
Well ID: W004 Well Name: SYN-Assam-Duliajan-004 Report Date: 2008-11-06 Day Number: 10 Start Depth: 3010 m End Depth: 3230 m Drilled Interval: 220 m Formation: Sandstone / Limestone Mud Weight: 13.41 ppg ROP: 12.4 m/hr RPM: 101 WOB: 30.7 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 220 m of section from 3,010 m to 3,230 m (22.5 hr on bottom). - Deviation survey taken; hole angle within tolerance.						
Geological observations. Formation change crossed during the tour: Sandstone to 3,180 m, Limestone to 3,590 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.41 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status. Depth 3,230 m in Sandstone. Drilling ahead. No non-productive time this tour.						

DAILY DRILLING REPORT — DAY 11

W004 · 07-Nov-2008

Well ID: W004

Well Name: SYN-Assam-Duliajan-004

Report Date: 2008-11-07

Day Number: 11

Start Depth: 3230 m

End Depth: 3390 m

Drilled Interval: 160 m

Formation: Limestone

Mud Weight: 13.93 ppg

ROP: 7.6 m/hr

RPM: 92

WOB: 34.7 klbf

Events: Mud Loss at 3237 m (Medium)

NPT: 7.4 hours

Operations performed.

• Drilled 160 m of Limestone from 3,230 m to 3,390 m (15.1 hr on bottom).

• Made connections and circulated bottoms-up at intervals; shakers monitored.

• Mud Loss handled at 3,237 m — 7.4 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,180–3,590 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 13.93 ppg after treatment.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0034	3,237	Limestone	Mud Loss	Medium	7.4	Partial non-productive time; drilling suspended while the well was treated and conditioned.

End-of-day status.

Depth 3,390 m in Limestone. Drilling ahead. Non-productive time this tour 7.4 hr.

DAILY DRILLING REPORT — DAY 12	W004 · 08-Nov-2008
Well ID: W004 Well Name: SYN-Assam-Duliajan-004 Report Date: 2008-11-08 Day Number: 12 Start Depth: 3390 m End Depth: 3590 m Drilled Interval: 200 m Formation: Limestone Mud Weight: 14.18 ppg ROP: 6.9 m/hr RPM: 89 WOB: 36.8 klbf Events: None NPT: 0.0 hours	
Operations performed. - Drilled 200 m of Limestone from 3,390 m to 3,590 m (16.5 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Pulled out of hole, ran 7" casing to 3,590 m and cemented as programmed (Tail 15.8 ppg, TOC 1,860 m). Waited on cement and pressure tested. - Reached driller's total depth of 3,590 m. Circulated hole clean, logged and secured the well.	
Geological observations. Cuttings throughout the interval are limestone, consistent with the 3,180–3,590 m unit logged in the offset wells of this cluster.	
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 14.18 ppg, properties held within programme; no losses or gains recorded.	
Equipment / operational observations. BOP drill carried out with the crew; response time satisfactory.	
Events reported this day. None. Operations proceeded within the planned parameter envelope.	
End-of-day status.	